

Pierce Lonergan

Software Engineer III | Data Engineering and Applied Machine Learning

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SUMMARY

Software Engineer III at JPMorganChase, four years and two promotions in Consumer & Community Banking data engineering, building the high-throughput Kafka and Spark infrastructure behind the lakehouse and the schema-evolution standards that keep it safe to change. Maintains three open-source packages on PyPI and Maven Central covering semantic schema matching, nested-record flattening, and governed identifier expansion, each with a reproducible benchmark.

TECHNICAL SKILLS

Streaming and Big Data: Apache Kafka, Apache Spark (Structured Streaming), Apache Avro, Project Reactor, Cassandra, schema evolution, change data capture (Debezium), exactly-once processing, transactional outbox

Lakehouse and Warehouse: Apache Iceberg, Snowflake, Parquet, partitioning strategies, data modeling, incremental sync

Applied ML and Retrieval: Retrieval-augmented generation, hybrid dense and BM25 retrieval, sentence embeddings, cross-encoder reranking, HNSW vector indexes, ONNX Runtime, int8 quantization, benchmark construction and evaluation design

Data Governance: Canonical catalogs, semantic schema matching, entity resolution, lineage, data-quality validation, data contracts, schema registry

Cloud and Platform: AWS (S3, EMR, MSK, Glue, Kinesis, Lambda), Docker, CI/CD (GitHub Actions), PostgreSQL, Spring Boot and Spring Modulith, hexagonal architecture

Languages: Python, Java, Scala, SQL, Groovy, Bash

PROFESSIONAL EXPERIENCE

JPMorganChase | Consumer & Community Banking, Data Engineering

Columbus, OH

Software Engineer III (Feb 2026 - Present) · Associate Software Engineer (Jan 2024 - Feb 2026) · Software Engineer (Apr 2022 - Jan 2024)

- Architected the reusable streaming-pipeline layer the team builds on: declarative source, sink, and transform factories, forward-compatible schema evolution, generic recursive flattening for arbitrarily nested payloads, and lifecycle concerns solved once and reused everywhere (checkpointing, error handling, graceful shutdown). Team throughput moved from roughly 2 to 30 production pipelines per month in the two quarters after adoption; the framework is mine, the throughput number is the team's.
- Shipped data-governance automation, now a production internal product, that replaced blank-page hand-mapping by data stewards with machine-proposed mappings they confirm or correct, cutting weeks from approval cycles (product-owner estimate, not an instrumented baseline). Drove schema-evolution standards, semantic schema matching, and entity resolution against a canonical catalog.
- Built high-performance Spark ETL with asynchronous processing and parallelization and integrated Kafka for near-real-time streaming; earlier, modernized big-data systems with reactive paradigms (Project Reactor), pairing Kafka messaging with Cassandra reconciliation, and migrated high-throughput systems to federated AWS.
- Placed 4th in JPMorganChase's global internal hackathon; reviewing architects projected roughly 80 engineer-hours per month saved if the prototype were productionized.

OPEN-SOURCE AND PUBLISHED WORK

Personal projects, built solo with an agent-assisted development loop. Architecture, invariants, benchmark design, and test strategy are mine; every figure below regenerates from a committed artifact.

nexus-matcher | semantic schema-field matching, Python

PyPI v2.2.0

- Maps Avro, JSON Schema, and SQL DDL fields onto data-dictionary entries by meaning rather than string equality: P@1 0.581, Recall@10 0.878, 364 fields/sec on CPU across 688 labelled pairs (BIRD-SQL 361, OMOP CDM v5.4 327).
- Kept the auto-approve gate conservative: at the 0.87 threshold it approves 85 of 688 fields (12.4%) at 95.3% precision and routes the rest to a human. The 22.9 MB wheel bundles its own int8 ONNX encoder, so it installs air-gapped with no model download and no torch.
- Retracted an earlier P@1 of 0.715 after tracing it to token-identity leakage in the OMOP split, and published the retraction in the repository README.

nexus-piercer | nested JSON and Avro flattening for Spark, Java 17

Maven Central

- Flattens deeply nested JSON and Avro into Spark-ready columns and reconstructs Avro records from flat rows: the nested-payload problem production ingestion hits, solved as a library. Published as `io.github.pierce-lonergan:nexus-piercer` with GPG-signed jar, sources, and javadoc per release, plus four install routes including a fully offline build.

acronymkit | governed identifier and abbreviation expansion, Python

PyPI v0.3.0

- Expands schema identifiers against a caller-supplied catalog and reports whatever the catalog cannot account for as unknown rather than guessing, which is what makes it safe for governance use. Precision 92.07%, recall 76.99%, F1 83.85% on the MED1250 benchmark; a claims gate fails CI when a documented figure cannot be traced to a recorded benchmark run.

momentum-x-research | research-methodology program, Python

Public repository

- Ran 311 documented experiments across 39 hypothesis families and 33 registered trials, certified zero edges, and published the null result: buy-and-hold SPY returned 14.91% per year over the same 2016 to 2026 window, above the program's own target.
- Built the evaluation stack that makes the null trustworthy: a hash-chained tamper-evident trial registry raising the promotion bar as the trial count grows (deflated Sharpe), hash-frozen pre-registrations with executable fixtures, death dates, and a day-clustered block bootstrap adopted after measuring within-day intraclass correlation at 0.09 to 0.35. Voided its own first passing result after review caught the runner running a 1-day horizon against a spec frozen at 21 days.

EDUCATION

The Ohio State University | B.S. Biochemistry, minor in Computer & Information Science

Aug 2016 - Dec 2021

Best Data Visualization, ASA DataFest 2021